Journal of Banking & Finance 55 (2015) 1–8

Contents lists available at ScienceDirect

Journal of Banking & Finance

journal homepage: www.elsevier.com/locate/jbf

Oil prices, US stock return, and the dependence between their quantiles ☆

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ARTICLE INFO

Article history:
Received 27 March 2014
Accepted 26 January 2015
Available online 20 February 2015

JEL classification:
G10
C01
C14
G14

Keywords:
Oil prices
Stock return
Local linear regression
Quantile regression

ABSTRACT

In this article, we examine the relationship between oil prices and US equities by proposing a novel *quantile-on-quantile* (QQ) approach to construct estimates of the effect that the *quantiles* of oil price shocks have on the *quantiles* of the US stock return. This approach captures the dependence between the distributions of oil price shocks and the US stock return and uncovers two nuance features in the oil–stock relationship. First, large, negative oil price shocks (i.e. low oil price shock quantiles) can affect US equities positively when the US market is performing well (i.e. at high US return quantiles). Second, while negative oil price shocks could affect the US stock market, the influence of positive oil price shocks is weak, which suggests that the relationship between oil prices on the US equities is asymmetric.
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1. Introduction

The hypothesis that oil prices exert important effects on equity returns has strong intuitive appeal but lack empirical support. Higher oil prices, for example, contribute to headline inflation that reduces real consumption.[1] They also lead to higher cost of production, which may constrain the economy and limit the number and types of investment opportunities available. According to the theory of asset pricing, a variable such as the price of crude oil that has plausible systemic influence on consumption (given wealth) or the investment opportunity set should affect the pricing of large stock aggregates. However, the empirical evidence corroborating the link between oil prices and stock markets has been tenuous at best (see, *inter alia*, Kling, 1985; Jones and Kaul, 1996; Chen et al., 1986; Huang et al., 1996),[2] with a number of recent papers even expressing doubts that oil prices could explain the behavior of stock returns.[3]

From a methodological perspective, there are various complexities in the relationship between oil prices and equities that make estimating this relationship potentially difficult with standard econometric techniques (such as ordinary least squares and quantile regressions). For example, equities may react to oil price shocks differently when the stock market is bearish than when it is bullish; the effect that large oil price shocks have on equities could be different from the effect that is due to smaller shocks; equities may also react asymmetrically to negative versus positive oil price shocks. Consequently, the effect of oil price shocks on equities may be heterogeneous, varying when market conditions and the nature of the oil price shocks differ.

Focusing on US equities, we re-investigate the effect that oil price shocks have on the US stock market through an empirical

☆ We are grateful to the editor Professor Carol Alexander and to the two anonymous referees for their very insightful comments and suggestions. We also like to thank Karim Chalak, James Hamilton, Lutz Kilian, Georg Strasser and Zhijie Xiao for helpful discussions.

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[1] Therefore, if today's higher oil prices are indicative of lower levels of real future consumption, agents might be more willing to accept a lower rate of return on financial assets in order to smooth consumption and transfer wealth to future periods of anticipated scarcity.

[2] For example, while some have found that oil price inflation can lead to stock market declines (e.g. Kling, 1985; Jones and Kaul, 1996), others have shown that the link between oil and stock prices is tenuous at best (e.g. Chen et al., 1986; Huang et al., 1996).

[3] For example, Wei (2003) finds no evidence that the U.S. stock market decline in 1974 is due to the oil price increase during that year.

http://dx.doi.org/10.1016/j.jbankfin.2015.01.013
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framework that can account for the above characteristics if they exist. In particular, we develop a new econometric approach that is suitable for estimating the effect of oil price shocks such that this effect may be contingent on (1) the performance of the US stock market, and (2) the sign and size of these shocks. This is achieved by incorporating two features into our modeling approach. First, we model the quantile of the US stock return as the dependent variable, since it is informative about how well or poorly the US stock market is performing. For example, the 98th percentile US return is indicative of a rapidly improving US stock market but the 2nd percentile US return is associated with a market that is strongly contracting. Second, we model the quantile of oil price shocks as an explanatory variable as this quantile contains information about both the sign and size of these shocks. For example, the 2nd percentile oil price shock is a large, negative shock, and in comparison, the 10th percentile shock is a smaller negative shock. Likewise, the 98th percentile shock is a large, positive shock while the 90th percentile shock is a smaller positive shock. Recognizing that the quantiles contain information about the state of the US stock market and the sign and size of the oil price shocks, we estimate the effect that the *quantile* of oil price shocks has on the *quantile* of the US stock return so as to examine the relationship between US equities and oil prices that pertains to different states of the US stock market and to oil price shocks of different signs and sizes. We call this approach of regressing a quantile on another quantile the *quantile-on-quantile* (QQ) approach.

The QQ approach is able to uncover some features in the relationship between US equities and oil prices that are not apparent if conventional approaches such as OLS or quantile regression are used. For example, using OLS regression, we find that the effect of oil price shocks on the US stock return is very weak. Similarly, using quantile regression, we find that oil price shocks have almost no bearing on the US return quantile. However, when the QQ approach is implemented, we find certain characteristics about the link between US equities and oil prices that elude both OLS and quantile regressions. For example, the QQ approach clearly demonstrates that *large*, *negative* oil price shocks (i.e. low oil price shock quantiles) can affect US equities positively when the US stock market is performing well (i.e. at high return quantiles). Moreover, it shows that the effect of positive oil price shocks is always weak, although negative oil price shocks could affect US equities as mentioned. This implies that the US equities-oil price relationship is contingent on the specific quantile of oil price shocks as well as the quantile of the US stock return, and thus, the sign and size of the oil price shocks and the state of the US stock market.

In pursuing this topic, there are two key issues that require attention. Firstly, oil prices do not determine stock returns exogenously, since equity and commodity prices may be driven by the same economic forces (e.g. stronger aggregate demand could drive up both oil and stock prices).[4] Therefore, this paper focuses on oil price shocks instead. To obtain oil price shocks that are not confounded by reverse causality or the presence of common factors, the literature has turned to structural vector autoregressions (SVAR) to model the channels of causality directly. For instance, through an SVAR approach, Kilian (2009) and Kilian and Park (2009) emphasize the importance of distinguishing between oil price shocks that are caused by actual oil production shortfalls or by shifts in precautionary demand for crude oil that reflect increased concerns about future supply shortages. In particular, they find that demand-led oil price shocks, but not supply driven shocks, are mainly responsible for explaining the link between US equities and oil prices. Hence, this motivates us to identify oil price shocks as shocks to the demand for oil in our empirical analysis.

Secondly, the QQ approach is developed by combining the techniques of quantile regression and nonparametric estimation. To tackle the "curse of dimensionality" problem associated with nonparametric models, we adopt a convenient dimension reduction technique known as local linear regression. Local linear regression involves taking a linear approximation of the unknown regression function around a chosen value of some explanatory variable (which yields the local linear model), then weighting all observations in the neighborhood of this choice with a kernel to construct a weighted objective function. In the context of ordinary least squares, local linear regression was first proposed by Cai et al. (2000) as a parsimonious way to model the conditional mean nonparametrically.[5] In this paper, we draw inspiration from Cai et al. (2000) (see, also, Cai and Xiao, 2012) by developing the QQ approach through the combination of quantile regression and local linear regression. This will allow us to estimate the dependence between the quantiles of the US stock return and oil price shocks for the following reason: if we specify the US stock return as the dependent variable, we may estimate its conditional quantile using quantile regression; if we use local linear regression to model the US stock return on oil price shocks, we may choose a specific oil price shock to examine the local effect that it has on the US stock return. Therefore, by (1) using local linear regression to choose some τ-quantile of oil price shocks and (2) implementing a θ-quantile regression with the US stock return as the dependent variable, we may then estimate the local effect that the τ-quantile of oil price shocks has on the θ-quantile of the US stock return.

Our paper is related to the literature within the area of portfolio management where it is known that the dependence (or correlation) between various classes of assets may be distribution specific. For instance, it is known that stock markets are more strongly correlated when they are bearish (Longin and Solnik, 2001), and therefore, the strength with which these markets are linked may vary with how well they are performing. Likewise, stronger tail dependence has also been observed between exchange rates (Patton, 2006; Bouyè and Salmon, 2009) and in the case of stocks and bonds, their relationship may change drastically during bear markets due to "flight to quality" away from equities (Guidolin and Timmermann, 2005). Viewing oil as an asset class, our work complements the literature by offering some evidence that the level of dependence between commodities (in this case, oil) and equities could be distribution (i.e. quantile) specific as well.

Our paper is also related to an emerging literature that seeks to provide a broad perspective on dependence by modeling the relationship between quantiles. In their seminal work, Ma and Koenker (2006) show that a triangular system of equations can be rewritten as a QQ-type framework that expresses the dependence between quantiles. A key methodological contribution of this paper is that unlike Ma and Koenker (2006), which requires a triangular system of equations to construct a QQ model, our approach can be implemented without triangular structures. In other words, our approach is applicable to "single equation" regression frameworks,[6] which implies that even for the most basic empirical models, it can be applied to estimate the general structure of dependence by relating the distribution of the dependent variable to the distribution of the regressor.

For the remainder of the paper, Section 2 introduces the QQ model. Section 3 discusses our data sources and Section 4 docu-

[4] Hamilton (1983) examines the correlation between oil price change and gross national product growth and finds that exogenous oil price changes had been responsible for the US recessions prior to 1973. For more discussions, see Mork (1989).

[5] By having the coefficients in the regression as unknown functions of some 'smooth variable', this so-called 'functional coefficient model' can achieve modeling flexibility and estimating feasibility simultaneously.

[6] For panel data regression, a pre-regression transformation to remove fixed effects in the data may be necessary before quantile regression is applied. See Canay (2011).

ments the relationship between oil price shocks and the US stock return using the QQ approach. Then, Section 5 performs a simple check to see if the QQ approach is valid. Section 6 concludes.

2. Methodology

In this paper, we examine the relationship between the US stock return and oil price shocks. While data on the US stock return is available,[7] the oil price shock has to be estimated first. In this paper, we adopt a two-step estimation procedure. In the first step, we employ the structural vector autoregression approach of Kilian and Park (2009) to uncover the structural shocks in oil prices. In the second step, we exploit these plausibly exogenous shocks to estimate the relationship between oil prices and US equities.

2.1. Step 1: Identification of the oil price shocks

The first step of the methodology concerns the identification of oil price shocks. There are two issues to consider here. First, some existing models have been constructed under the assumption of strict exogeneity of oil prices. This is implausible as there are economic fundamentals that could drive both stock returns and oil prices jointly, which makes separating cause from effect in this case difficult (see, e.g. Barsky and Kilian, 2002, 2004).[8] Second, even if reverse causality can be dealt with, it is important to distinguish between demand versus supply shocks to the crude oil market. For example, Kilian and Park (2009) argue that demand and supply shocks could differ in their effects on the U.S. economy and on the real price of oil (see, also, Kilian, 2008, 2009).[9] In fact, Kilian (2009) and Kilian and Park (2009) have found that oil price changes caused by oil demand shocks have significant effects on the US stock market, but the response of US equities to oil supply shocks is usually weak. Given that it is mainly oil demand shocks that matter, it is therefore important for us to first extract the demand driven portion of the innovation in oil prices.

To do so, we follow Kilian and Park (2009) and construct an SVAR model to identify the exogenous shocks to the real price of oil that are driven by demand. Kilian and Park (2009) call this SVAR system the "Global Oil Market Block", which is a block of three variables ordered according to the percent change in global crude oil production, followed by a measure of real economic activity that proxies for global aggregate demand, then the real price of crude oil. Letting $\mathbf{z}_t$ be the vector of these three variables, this SVAR can be expressed as

$$A_0\mathbf{z}_t = \alpha + \sum_{i=1}^{p} A_i\mathbf{z}_{t-i} + \boldsymbol{\epsilon}_t \tag{1}$$

where $\boldsymbol{\epsilon}_t$ is the vector of structural shocks that are assumed to be serially and mutually uncorrelated.[10] If $\mathbf{e}_t$ denotes the vector of reduced-form VAR residuals, the structural shocks and the VAR residuals are linked together through $\mathbf{e}_t = A_0^{-1}\boldsymbol{\epsilon}_t$. Consequently, to identify $\boldsymbol{\epsilon}_t$, we need to impose some structure on A_0^{-1}. We follow Kilian and Park (2009) by postulating a triangular structure on A_0^{-1} such that the link between the reduced form errors ($\mathbf{e}_t$) and the structural shocks ($\boldsymbol{\epsilon}_t$) is expressed as

$$\mathbf{e}_t = \begin{pmatrix} e_{1t} \\ e_{2t} \\ e_{3t} \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ a_{21} & 1 & 0 \\ a_{31} & a_{32} & 1 \end{bmatrix}^{-1} \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \\ \epsilon_{3t} \end{pmatrix} \tag{2}$$

where $\epsilon_{1t}, \epsilon_{2t}$ and ϵ_{3t} are the structural shocks to global oil production, global real activity, and real price of oil respectively. The triangular structure of A_0 is in turn motivated by three identifying restrictions that will pin down ϵ_{3t} specifically as the demand driven shock to the real price of oil.[11]

The first identifying restriction is based on the idea that short run adjustments to oil production may be costly, hence, the supply function of crude oil supply is inelastic within a month. Consequently, global oil production will not be affected by shocks to both global real activity and the real price of oil within a month, which implies that the reduced form error e_{1t} is equal to ϵ_{1t}. The second identifying restriction relies on the observation that global real activity is sluggish, and hence within a month, will not respond to variations in the real price of oil driven by demand shocks. In this case, the structural shock to global real activity (ϵ_{2t}) could be identified from the reduced form error e_{2t} once we have "partialled out" from it, the global oil production shock (ϵ_{1t}). The final restriction assumes that demand driven shocks to the real price of oil (ϵ_{3t}), which is the key variable of interest, are oil price shocks that are explained neither by oil production shocks nor shocks to global real activity. This assumption implies that the demand driven oil price shocks could be obtained from the reduced form error e_{3t} after removing from it, the shocks to crude oil supply (ϵ_{1t}) and global aggregate demand (ϵ_{2t}).

One of the fundamental insights of Kilian and Park (2009) is the inclusion of global real activity that is ordered as the second variable in the SVAR. According to them, the implication of not doing so is that a positive innovation to the global business cycle will stimulate the U.S. economy directly and drive up the price of oil at the same time.[12] Consequently, a positive shock to global aggregate demand may lead to an increase in both oil and equity prices (after controlling for crude oil production), and this positive co-movement will mitigate any negative effect that oil price shocks might have on the return on equities. In other words, without the second equation in the SVAR, the "shock" to oil prices will contain information about global aggregate demand that affects oil and equity prices in the same direction, and this correlation driven by global business cycles will confound the causal effect that runs from oil to equities.

With the ordering of crude oil production first, followed by global real economic activity, and then the real price of oil, ϵ_{3t} is the innovation in oil prices that is not related to shocks to crude oil production and global aggregate demand. Kilian and Park (2009) call this the demand shock to the crude oil market, but this statement needs clarification because global aggregate demand shocks, which presumably affect demand for oil as well, have already been "partialled out" from ϵ_{3t} based on the ordering of the SVAR. According to Kilian and Park (2009), ϵ_{3t} is intended to capture the shift in precautionary demand for crude oil that arises from increased uncertainty about shortfalls of expected supply relative to expected demand. This precautionary demand will

[7] As suggested in Kilian and Park (2009), the stock return is constructed from the CPI inflation rate and the log returns on the Center for Research in Security Prices (CRSP) value-weighted market portfolio.

[8] For example, Barsky and Kilian (2002) argue that the worsening stagflation in the 1970s and the simultaneously sharp increase in oil prices may not be entirely causal. In other words, the rise in oil prices during that period could in turn reflect an endogenous response of oil prices to macroeconomic conditions and ultimately by monetary policy.

[9] In particular, since the relative importance of these shocks could change over time, they argue that regressions of stock returns on oil price innovations could be biased toward finding the absence of a relationship or relationships that are unstable over time (Sadorsky, 1999).

[10] In the actual application, we use a 3-lag VAR. This is suggested by various lag selection criteria, such as the Akaike Information Criterion, Schwarz Criterion, the Hannan–Quinn Criterion, and the Forward Prediction Error.

[11] Kilian and Park (2009) interpret demand driven shocks to the real price of oil as shifts in precautionary demand for oil arising from increased concerns about the availability of future oil supplies.

[12] Although the resulting increase in the price of oil has a dampening effect on economic activity, it may not be enough to overcome the positive *direct* effect that global aggregate demand has on the US economy.

affect the demand for oil inventory holdings as a way of insuring against any unexpected interruption in oil supplies, which in turn will affect the price of oil. Therefore, because ϵ_{3t} is the shock in oil prices that is due to changes in the precautionary demand for oil, it is in this sense that Kilian and Park (2009) call it the demand driven oil price shock.

2.2. Step 2: The local linear quantile regression

For convenience, let us rephrase "demand driven oil price shocks" simply as "oil price shocks" and let us denote ϵ_{3t} as Oil_t. After obtaining an estimate of Oil_t from Step 1, we utilize a nonparametric approach to examine the effect that large (or small) oil price shocks might have on the distribution of the US stock return. To do so, we construct a model that relates the quantile of the US stock return with the quantile of oil price shocks, so that the relationship between oil prices and US equities could vary at different points in their respective distributions. Letting the θ superscript denote the quantile of the US stock return, we first postulate a model for the θ-quantile of the US stock return (r_t) as a function of oil price shocks as[13]

$$r_t = \beta^\theta(Oil_t) + \alpha^\theta r_{t-1} + v_t^\theta, \tag{3}$$

where v_t^θ is an error term that has a zero θ-quantile. We allow the link function $\beta^\theta(\cdot)$ to be unknown, which in our view is a sensible assumption given that we do not have a prior on how the US stock return and oil price shocks are related.[14] To study the relationship between θ-quantile of the US stock return and the τ-quantile of oil price shocks, denoted by Oil^τ, we examine Eq. (3) in the neighborhood of Oil^τ. Given that $\beta^\theta(\cdot)$ is unknown, we linearize this function by taking a first order Taylor expansion of $\beta^\theta(\cdot)$ around Oil^τ, which leads to

$$\beta^\theta(Oil_t) \approx \beta^\theta(Oil^\tau) + \beta^{\theta\prime}(Oil^\tau)(Oil_t - Oil^\tau). \tag{4}$$

One interesting feature about Eq. (4) is the fact that its parameters $\beta^\theta(Oil^\tau)$ and $\beta^{\theta\prime}(Oil^\tau)$ are doubly indexed in θ and τ. To see why, notice that Oil^τ, the τ-quantile of oil price shocks, is a function of τ alone. Since $\beta^\theta(Oil^\tau)$ and $\beta^{\theta\prime}(Oil^\tau)$ are functions of θ and Oil^τ, and since Oil^τ is a function of τ, this implies that $\beta^\theta(Oil^\tau)$ and $\beta^{\theta\prime}(Oil^\tau)$ are both functions of θ and τ. From this perspective, let us redefine $\beta^\theta(Oil^\tau)$ and $\beta^{\theta\prime}(Oil^\tau)$ as $\beta_0(\theta,\tau)$ and $\beta_1(\theta,\tau)$ respectively. Then, rewrite Eq. (4) as

$$\beta^\theta(Oil_t) \approx \beta_0(\theta,\tau) + \beta_1(\theta,\tau)(Oil_t - Oil^\tau), \tag{5}$$

and substitute Eq. (5) into Eq. (3) to obtain

$$r_t = \underbrace{\beta_0(\theta,\tau) + \beta_1(\theta,\tau)(Oil_t - Oil^\tau) + \alpha(\theta) r_{t-1}}_{(*)} + v_t^\theta, \tag{6}$$

where we have used the notation $\alpha(\theta) \equiv \alpha^\theta$. Part $(*)$ of Eq. (6) is the θ conditional quantile of the US stock return. However, unlike a standard conditional quantile function, this expression captures the relationship between the θ-quantile of the US stock return and τ-quantile of oil price shocks, given that β_0 and β_1 are doubly indexed in θ and τ. In this regard, Part $(*)$ of Eq. (6) can capture the overall dependence structure between the US stock return and oil price shocks through the dependence between their respective distributions.

To estimate Eq. (6), we replace Oil_t with its estimated counterpart $\widehat{Oil}_t$. We also replace Oil^τ with its empirical quantile of $\widehat{Oil}_t^\tau$. Then, we solve for

$$\min_{b_0,b_1} \sum_{i=1}^{n} \rho_\theta \left[r_t - b_0 - b_1(\widehat{Oil}_t - \widehat{Oil}^\tau) - \alpha(\theta) r_{t-1} \right] K\left(\frac{F_n(\widehat{Oil}_t) - \tau}{h} \right)$$

to obtain the estimates $\hat{\beta}_0(\theta,\tau)$ and $\hat{\beta}_1(\theta,\tau)$, where the function ρ_θ is the tilted absolute value function that gives the θ-conditional quantile of r_t as the solution. Because we are interested in the effect exerted locally by the τ-quantile of oil price shocks, we employ a Gaussian kernel $K(\cdot)$ to weight the observations in the neighborhood of $\widehat{Oil}^\tau$, based on bandwidth h. These weights are inversely related to the distance of $\widehat{Oil}_t$ from $\widehat{Oil}^\tau$, or more conveniently, the distance of the empirical distribution function

$$F_n(\widehat{Oil}_t) = \frac{1}{n} \sum_{k=1}^{n} I(\widehat{Oil}_k < \widehat{Oil}_t) \tag{7}$$

from τ, where τ is the value of the distribution function that corresponds with Oil^τ. In the empirical exercise below, we choose a plug-in bandwidth of $h = 0.05$, but it should be noted that the issues involving kernel regression apply here as well. In particular, a major tension in kernel regression is the choice of the bandwidth. If the chosen bandwidth is small (i.e. if we undersmooth), the bias of the estimates would be smaller but their variances would increase. Conversely, if the chosen bandwidth is large, the variance of the estimates would be smaller but their bias would increase.

3. Data and summary statistics

We employ monthly data from January 1973 to December 2007. The first step of our two-step procedure utilizes data on world crude oil production, the real crude oil price and a measure of global real activity.[15] Data on the first two variables are straightforward to obtain from the U.S. Department of Energy. Following Kilian and Park (2009), information on global real activity is constructed using an index related to a panel of cargo ocean shipping freight rates, based on the argument global aggregate demand is well reflected by strong global demand for industrial commodities, which in turn affects the demand for shipping and thus shipping rates. The second step of our estimation procedure utilizes the US stock return, which following Kilian and Park (2009), is the real return on the US stock market based on the log returns on the Center for Research in Security Prices (CRSP) value-weighted market portfolio adjusted for CPI inflation.

Table 1 presents the summary statistics of two key variables in this paper: the US stock return and oil price shocks. Two points are worth mentioning here. First, the monthly mean and median of the US stock return is about 0.01 (or 1 percent). This translates into an annualized return of about 12 percent per year. Second, with a correlation coefficient of −0.09, the US stock return is very weakly related to oil price shocks. Therefore, it is not surprising that OLS and quantile regressions, as we will see in Section 4, are unable to say much about the link between the US stock return and oil price shocks.

The QQ approach proposed in this paper focuses on how the US stock return and oil price shocks are related at their various quantiles. To get a sense of what these quantiles look like, Fig. 1(a) first plots the quantile of the US stock return and shows that the return below the 40th percentile is negative. Hence, the lower quantiles of

[13] This framework is also known in the literature as the *partially varying coefficient model*. See Cai et al. (2000).

[14] In nonparametric econometrics, Eq. (3) belongs to the class of partial linear models that is popularized by the landmark contribution of Robinson (1988). Perhaps one of the most popular application of the partial linear framework is Levinsohn and Petrin (2003) in the estimation of production functions. Recent work in time series include Gao (2007), Gao et al. (2006) and Lu et al. (2009).

[15] We use the same data as Kilian and Park (2009) for these 3 variables.

Table 1
Summary statistics.

	US return	Oil price shock
Mean	0.01	0
Standard deviation	0.05	0.99
Minimum	−0.23	−4.67
Maximum	0.16	5.85
Correlation	−0.09	–

the US stock return are indicative of a bearish US market, and vice versa. Fig. 1(b) presents a similar plot for the oil price shocks. Because the oil price shocks are standardized, any shock that is above (below) the median is positive (negative). In other words, negative oil price shocks are shocks that are below the median, and vice versa.

4. Empirical results

As a preliminary exercise, we first explore what OLS and quantile regressions have to say about the effect that oil price shocks have on the US stock return. The result of the OLS regression is given by

$$r_t = \underset{(0.0023)}{0.0061} - \underset{(0.0023)}{0.0043}\, Oil_t + \underset{(0.0489)}{0.0387}\, r_{t-1}. \tag{8}$$

Eq. (8) suggests that the effect of oil price shocks is very weak, since it shows that negative oil price shocks as large as two standard deviations could only produce a 0.8 percentage point increase in the US stock return on average. Furthermore, this effect is statistically insignificant at the 5 percent level. Therefore, if OLS regression is used, the link between oil prices and US equities would seem tenuous.

Quantile regression (QR) offers more information than OLS regression does as it looks at the effect that oil price shocks have on the distribution of the US stock return. The model estimated by quantile regression is given by

$$\hat{r}_t^{\theta} = \hat{\gamma}_0(\theta) + \hat{\gamma}_1(\theta) Oil_t + \hat{\gamma}_2(\theta) r_{t-1}, \tag{9}$$

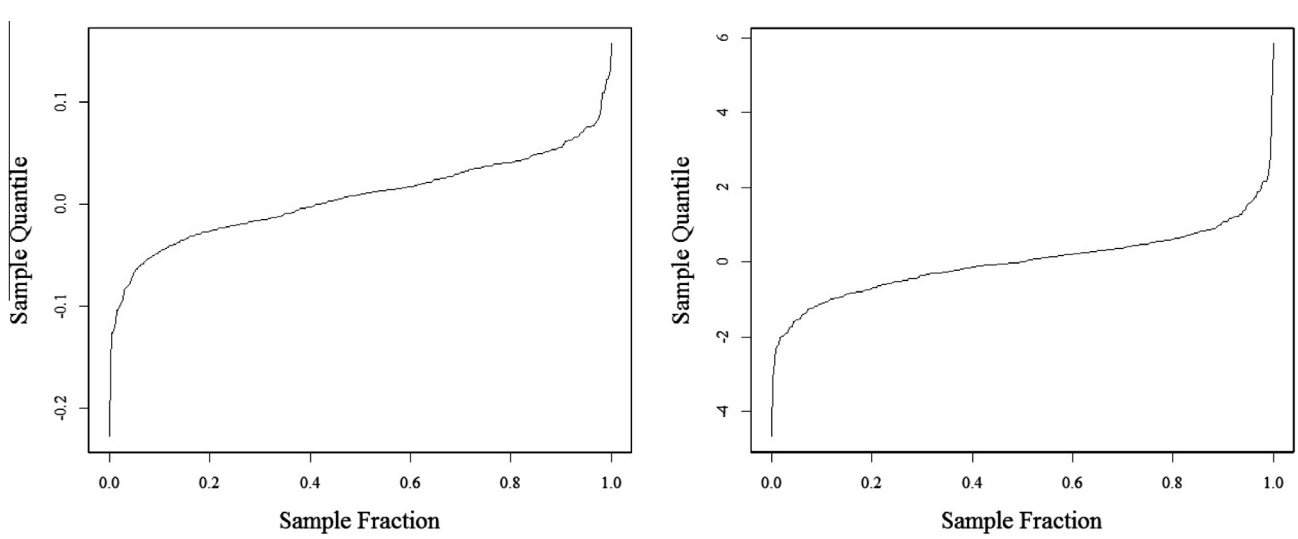
where $\hat{r}_t^{\theta}$ is the estimated θ-conditional quantile of the US stock return and the estimated parameters, $\hat{\gamma}_k(\theta)$ for $k = 0, 1, 2$, are functions of θ. Fig. 2(a) plots the intercept term $\hat{\gamma}_0(\theta)$ against θ together with the two standard deviation bands. It shows that $\hat{\gamma}_0(\theta)$ is increasing in θ, which implies that $\hat{\gamma}_0(\theta)$ is larger for the upper quantiles of the US stock return. This is not surprising as the intercept term reflects the "level" of the US stock return for given levels of Oil_t and r_{t-1}. Therefore, all things being equal, a larger US stock return (i.e. a higher return quantile) should have a larger intercept term.

From Eq. (9), the effect of oil price shocks on the θ-quantile of the US stock return is captured by $\hat{\gamma}_1(\theta)$. Fig. 2(b) plots $\hat{\gamma}_1(\theta)$ against θ and shows that $\hat{\gamma}_1(\theta)$ is mostly statistically insignificant across θ. Furthermore, not only is $\hat{\gamma}_1(\theta)$ close to zero, it also displays very little variation across θ. This implies that the effect of oil price shocks on the quantiles of the US stock return is weak, hence based on evidence from quantile regression, US equities do not appear to be tightly linked with oil prices.

One shortcoming of the QR approach is in its inability to capture dependence in its entirety. Specifically, even though the QR approach can estimate the heterogeneous relationship between the US stock return and oil price shocks pertaining to different US return quantiles, it overlooks the possibility that the nature of oil price shocks could also influence the way oil prices and equities are related. For example, if the effect of large, positive oil price shocks is different from the effect of large, negative shocks, the *same* US return quantile would react to positive versus negative oil price shocks asymmetrically. Unfortunately, such subtle economic behaviors cannot be captured easily by quantile regression.

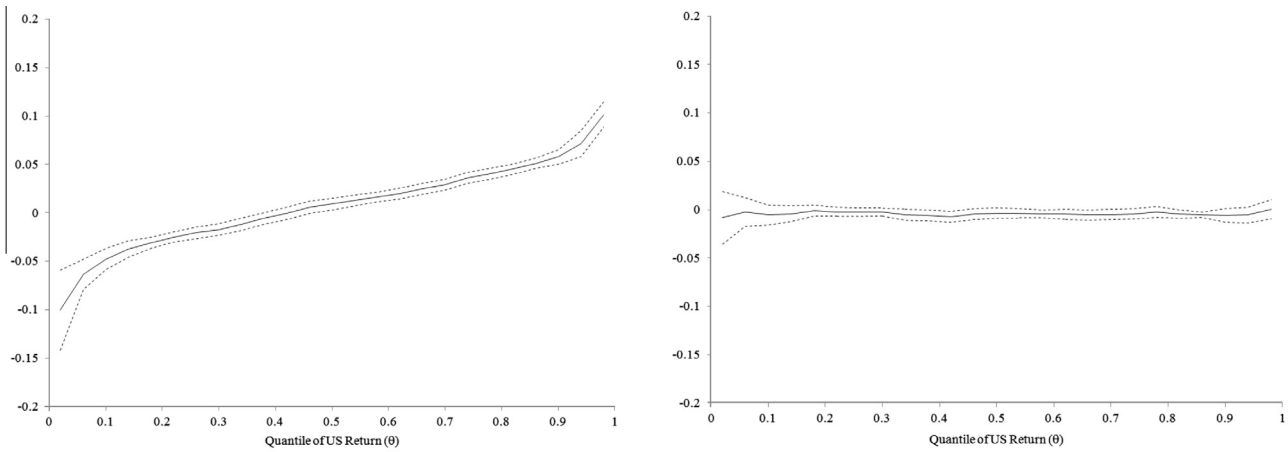
The QQ approach is developed for the purpose of providing a more complete picture of dependence. It is implemented by selecting a number of quantiles of oil price shocks (indexed by τ) and estimating the local effect that the τ-quantile of oil price shocks might have on the θ-quantile of the US stock return. According the QQ model expressed in Eq. (6), the overall dependence structure of oil prices and equities can be summarized by two parameters: $\beta_0(\theta, \tau)$ and $\beta_1(\theta, \tau)$. Being functions of θ and τ, these parameters may vary across different θ-quantiles of the US stock return and τ-quantiles of oil price shocks, thereby provide information how the US stock return and oil price shocks are related at their respective distributions.

Using the QQ approach, we first examine the relationship between the US stock return and oil price shocks that is reflected by $\beta_0(\theta, \tau)$, the intercept term of the US stock return equation in Eq. (6). Because $\beta_0(\theta, \tau)$ is indexed by τ, it may vary with different



(a) Quantile of US Stock Return (b) Quantile of Oil Price Shock

Fig. 1. Quantile plots.



(a) $\hat{\gamma}_0(\theta)$ (b) $\hat{\gamma}_1(\theta)$

Fig. 2. QR estimates.

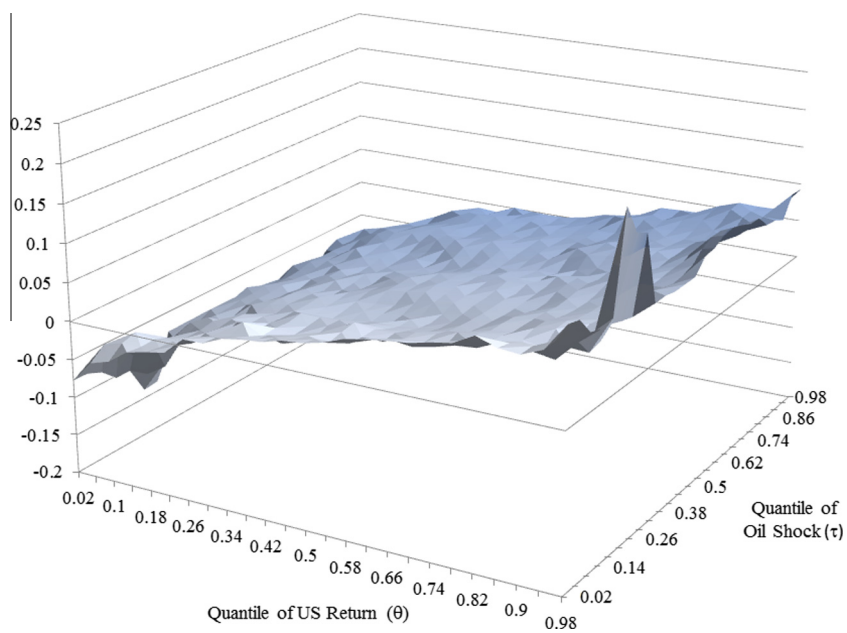


Fig. 3. QQ estimate, $\hat{\beta}_0(\theta,\tau)$.

τ-quantiles of oil price shocks, unlike the intercept term of the QR approach in Eq. (9) that is insensitive to τ. Fig. 3 plots the estimate of $\beta_0(\theta,\tau)$ (denoted by $\hat{\beta}_0(\theta,\tau)$) in the *z*-axis against the quantiles of US stock return and oil price shocks in the *x*–*y* axes. Based on Fig. 3, three things can be said about $\hat{\beta}_0(\theta,\tau)$.

Firstly, $\hat{\beta}_0(\theta,\tau)$ is increasing in θ. This implies that the intercept is larger at higher US return quantiles (i.e. higher θ), which is what we have observed before (see Fig. 2(a)). Secondly, depending on θ, $\hat{\beta}_0(\theta,\tau)$ may vary with τ, hence, across different τ-quantiles of oil price shocks. For example, around the upper US return quantiles (e.g. $\theta = 0.9$), $\hat{\beta}_0(\theta,\tau)$ exhibits an upward spike when the quantile of oil price shocks is below the forth decile (i.e. $\tau < 0.4$). This upward spike in $\hat{\beta}_0(\theta,\tau)$, which takes place in the vicinity of $\theta = 0.9$ and $\tau < 0.4$, shows that large negative oil price shocks can bolster the return on equities further when the US market is optimistic. Interestingly, when the market is performing less well, there is no evidence that negative oil price shocks can generate such positive responses in equities. For example, at its median, (i.e. $\theta = 0.5$), the US stock return responds very weakly to variations in τ. If the median US stock return represents a market that is neither bearish nor bullish, this implies that oil price shocks have little effects on US equities during non-extreme periods of market activity. These observations, therefore, suggest that the relationship between oil prices and US equities could depend on both the nature of oil price shocks and how well the US stock market is performing.

Finally, the manner in which $\hat{\beta}_0(\theta,\tau)$ behaves with respect to τ suggests that the effect of oil price shocks could be asymmetric. For instance, no matter which θ-quantiles of the US stock return we consider, the insensitivity of $\hat{\beta}_0(\theta,\tau)$ with respect to the right-tailed (i.e. $\tau > 0.5$) quantiles of oil price shocks demonstrates that the effect of positive oil price shocks on equities is weak. In contrast, when the US stock market is performing well, a large enough negative oil price shock (i.e. $\tau < 0.4$) could produce a positive impetus on the US stock return as we have just seen. Therefore, while nega-

tive oil price shocks may affect the US stock market, positive oil price shocks appear to be irrelevant.

Besides the intercept term $\beta_0(\theta,\tau)$, the effect of oil price shocks is also captured by the slope coefficient of the τ-quantile of oil price shocks, $\beta_1(\theta,\tau)$. Fig. 4 plots the estimate of $\beta_1(\theta,\tau)$ (denoted by $\hat{\beta}_1(\theta,\tau)$) at different values of θ and τ. Generally, $\hat{\beta}_1(\theta,\tau)$ tends to meander around zero, implying that there are large regions in the distributions of the US stock return and oil price shocks where these variables appear to be un-coupled. However, in the upper θ-quantiles of the US stock return, there are regions where $\hat{\beta}_1(\theta,\tau)$ is negative when τ is low. This suggests that negative oil price shocks (i.e. low τ) could further bolster the US stock market when it is performing well (i.e. high θ). Once again, Fig. 4 shows that positive oil price shocks have no bearing on equities in general, hence, the effect of oil price shocks is asymmetric.

5. Checking the QQ approach

In principle, what the QQ approach does is to "decompose" the QR estimates so that they are specific for different quantiles of the explanatory variable (e.g. oil price shocks). For example, the QR approach regresses the θ-quantile of the US stock return on oil price shocks, hence, the parameters in the QR model will be indexed by θ only (see Eq. (9)). The QQ approach, however, regresses the θ-quantile of the US stock return on the τ-quantile of oil price shocks, and as a result, its parameters will be indexed by both θ and τ. Therefore, the QQ approach should convey more information about the relationship between US equities and oil prices than the QR approach would, since the QQ approach, but not the QR approach, views this relationship as potentially heterogeneous across τ.

If the QQ approach does in fact "decomposes" the QR estimates as claimed, it will be possible to use the QQ estimates to recover the key features about the QR estimates; otherwise, the QQ approach is methodologically flawed. To recover the QR estimates from the QQ estimates, first, notice that the QR parameters are indexed by θ only. Hence, to construct parameters from the QQ model that are indexed by θ only, we summarize the estimated QQ parameters by averaging along τ:

$$\hat{\gamma}_0^*(\theta) \equiv \bar{\hat{\beta}}_0(\theta) = \frac{1}{S}\sum_{\tau}\hat{\beta}_0(\theta,\tau), \quad \hat{\gamma}_1^*(\theta) \equiv \bar{\hat{\beta}}_1(\theta) = \frac{1}{S}\sum_{\tau}\hat{\beta}_1(\theta,\tau) \tag{10}$$

where $S = 45$ is the number of points of the grid $\tau = [0.02, 0.04, \ldots, 0.98]$. Given that the QQ parameters in (10) are

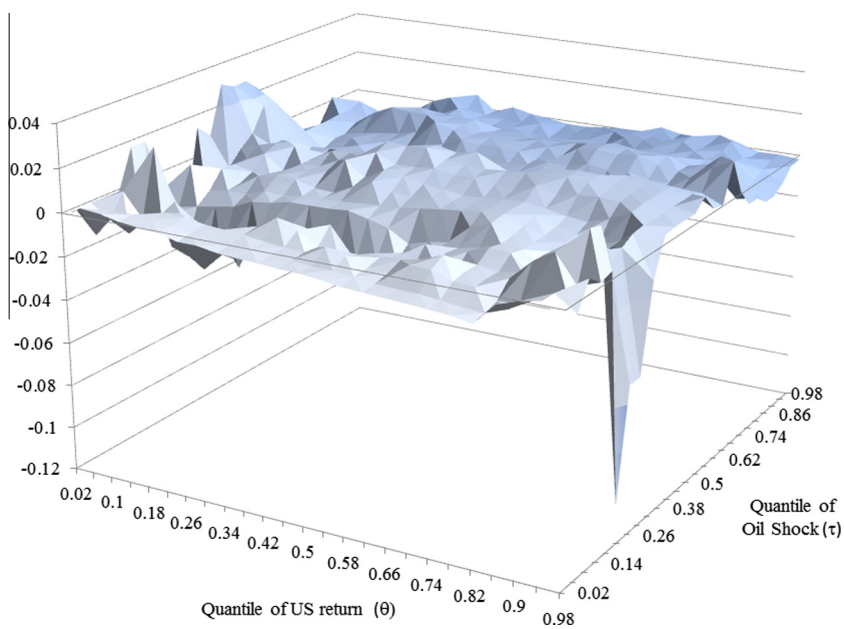
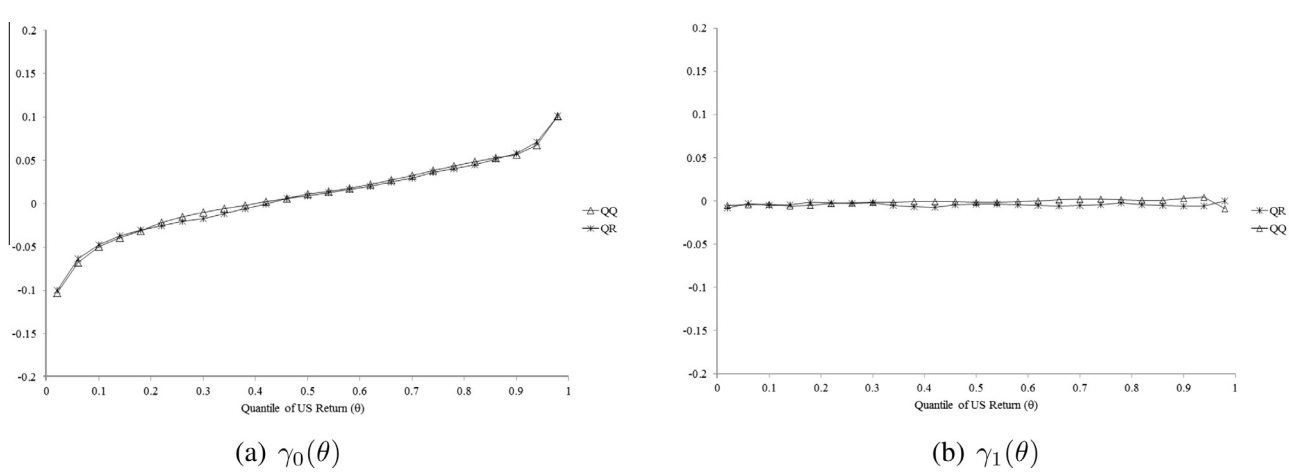


Fig. 4. QQ estimate, $\hat{\beta}_1(\theta,\tau)$.



(a) $\gamma_0(\theta)$ (b) $\gamma_1(\theta)$

Fig. 5. Comparison of the QR and the averaged QQ estimates.

now indexed by θ only, a heuristic way of checking if the QQ approach is valid is to compare these τ-averaged QQ parameters (i.e. $\hat{\gamma}_0^*(\theta)$ and $\hat{\gamma}_1^*(\theta)$) with the estimated QR parameters (i.e. $\hat{\gamma}_0(\theta)$ and $\hat{\gamma}_1(\theta)$).

Fig. 5 plots the QQ and QR estimates of $\gamma_0(\theta)$ and $\gamma_1(\theta)$. With respect to $\gamma_0(\theta)$, not only is the QQ estimate increasing in the θ-quantile of the US stock return, it is also nearly identical with the QR estimate (see Panel (a)). Similarly, the QQ and QR estimates of $\gamma_1(\theta)$ are both very close to zero across all θ (see Panel (b)). Therefore, Fig. 5 provides a simple validation of the QQ approach by showing that the key features of the QR estimates can be recovered by summarizing the more disaggregated information contained in the QQ estimates.

6. Conclusion

In this paper, we propose a new estimation method, dubbed the QQ approach, to estimate the overall dependence structure between US equities and oil prices. A key advantage of the QQ approach is in its ability to model economic relationships more richly than is possible with OLS or quantile regression. For example, OLS regression is only able to estimate the effect that oil price shocks have on the *conditional mean* of the US stock return; quantile regression goes further by decomposing this effect on the conditional mean into effects on the *conditional quantile*. The QQ approach, in turn, generalizes the quantile regression approach by shedding light on how the *quantiles* of oil price shocks could affect the *conditional quantile* of the US stock return. Among the three estimation approaches, the QQ approach offers the most information possible about the relationship between US equities and oil prices. Indeed, through the QQ approach, we can gather some insights on how highly complex this relationship might be. For instance, we find that negative oil price shocks can improve the return of US equities when the US market is performing well, but the explanatory power of positive oil price shocks is always weak. Although this behavior is rather subtle, the QQ approach has the ability to uncover it quite straightforwardly.

To the best of our knowledge, our paper is among the very first to employ a QQ-type analysis in macroeconomics and finance. While the dependence of quantiles may seem quite complex, our paper shows this is not difficult to estimate at all. In fact, our paper uses the insight that the combination of quantile and local linear regression – two rather standard procedures in the econometric literature – can deliver a powerful way to capture the dependence of covariates richly in the spirit of quantile dependence. Because our approach does not require triangular structures, unlike the QQ technique of Ma and Koenker (2006), it can be applied virtually to any "single equation" models in macroeconomics and finance, such as models that relate the effect of monetary policy shocks on the stock market (e.g. Cover, 1992; Senda, 2001) or on output growth (e.g. Conover et al., 1999; Chen, 2007). Therefore, although we are interested in equities and oil prices, the QQ approach developed here is not constrained for this application. Instead, it is a general methodology that could be applied to estimate and detect complex characteristics such as nonlinearities and asymmetries that standard econometric frameworks may not be equipped to handle.

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